

Quantification of Protein Mixture in Chromatographic Separation Using Multi-Wavelength UV Spectra

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*In therapeutic protein production, the protein purification with chromatographic processes is of high importance in separating the qualified proteins from the impurities for consistent product quality. Therefore, to aid real-time monitoring of the protein purification processes, various kinds of methodologies have been proposed until now. However, the majority of them still rely on the use of a single ultraviolet (UV) absorbance or the utilization of expensive and time-consuming instruments, thus requiring a simple, fast, and cost-effective methodology for protein quantification. In this study, the feasibility of using multiwavelength UV spectroscopy was investigated as an alternative tool for the real-time monitoring of the protein mixtures in protein purification. To this end, three different proteins were selected as a model system for the protein mixture, and the multivariate UV spectra were analyzed to construct the reliable quantification models for different proteins of interest. By using various chemometrics tools, such as partial least squares (PLS), the validity of estimating the protein concentration from the UV spectra of the mixture samples was rigorously analyzed with their prediction performance, and the results indicated that the multiwavelength UV spectra had sufficient sensitivity and accuracy to estimate the protein concentrations in mixture, demonstrating its usefulness for the rapid quantification of the protein mixtures in protein purification. © 2013 American Institute of Chemical Engineers *Biotechnol. Prog.*, 29:664–671, 2013*
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Introduction

Among the several unit operations in the typical biopharmaceutical manufacturing processes, the protein purification step is generally of high importance in obtaining the qualified target protein from the impurities, and the chromatographic separation system has been often employed for this purpose by separating protein mixtures according to their differences in physico-chemical properties.^{1–3} Owing to its importance in biopharmaceutical manufacturing as well as other related industries, many efforts have been made until now to develop an efficient chromatographic process.^{4–7} However, despite these advances, the protein purification is often suffered from the instabilities due to the complex interactions among the protein mixtures, lot-to-lot variability in resin materials and others.⁸ Therefore, to minimize the subsequent impacts on the final product quality of the therapeutic proteins, a reliable monitoring tool for the

chromatography operation is highly necessitated,^{9,10} which is especially in line with the recent initiatives of process analytical technology (PAT) and quality by design (QbD) to promote the improved understanding of the underlying pharmaceutical manufacturing processes through the timely measurements of critical process parameters or quality attributes.^{11–13}

Currently, most conventional methods to quantify the proteins in the effluents of chromatography are based on the monitoring of ultraviolet (UV) absorbance at an univariate wavelength (i.e., 280 nm),¹⁴ since it is very fast, simple, and easy to implement. However, it cannot often differentiate the target protein from other species or impurities, thus having a limited capability to aid reliable real-time pooling decision during the chromatographic operations. To resolve this limitation, various kinds of PAT tools have been proposed until now, including online high performance liquid chromatography (HPLC),¹⁴ ultra performance liquid chromatography (UPLC),¹⁵ and tryptophan fluorescence.¹⁶ However, these approaches often require expensive instruments, frequent maintenance and significant consumption of chemicals, thus demanding an alternative approach for real-time protein quantification in chromatographic systems.

Additional Supporting Information may be found in the online version of this article.

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Under these circumstances, the use of spectroscopic measurements together with chemometrics techniques can be a viable tool in estimating the physico-chemical properties of the protein mixtures in a simple, fast, and cost-effective manner.^{17,18} Chemometrics-based multivariate statistical methods, such as principal component analysis (PCA) and partial least squares (PLS) have been frequently used to analyze the full spectra of biological samples (i.e., near-infrared or Raman spectra), and have demonstrated its usefulness in developing a novel PAT tool.^{19,20} Indeed, the applications of chemometrics in conjunction with spectroscopic measurements have become one of the key methodologies to enable the success of PAT in characterizing both critical process parameters and product quality attributes in pharmaceutical and biopharmaceutical manufacturing processes.^{21,22}

In this study, the feasibility of multiwavelength UV absorption spectra was investigated to estimate the protein concentrations in mixture as a potential real-time quantitative monitoring tool in protein purification, considering that it can be readily implemented in many of the existing chromatography systems with minimal modification and has been well-established techniques with maximal robustness and reproducibility.²³ In fact, several researchers have previously reported the successful applications of UV spectroscopy in analyzing the structural behavior of the protein or in quantifying the composition of proteins mixtures.^{24–27} However, the majority of them have still used a univariate or a few wavelengths of high importance (i.e., ratio of two absorbance, etc.), and thus could not take the full advantages offered by multivariate information contained in the full spectrum. Although a few researchers have recently analyzed the UV spectra of protein samples based on the approach of multivariate calibration,^{24,25} the general low sensitivity of UV absorbance for multicomponent protein samples with overlapping peaks requires a rigorous analysis of the multivariate spectra as well as a statistical validation of estimation performance to facilitate its wide acceptance in the quantitative monitoring of protein mixtures in chromatographic operations.

In this study, a ternary protein mixture of immunoglobulin G (IgG), bovine serum albumin (BSA), and β -lactoglobulin (BLG), having distinct molecular weight and chemical structure, was selected as a model system, and a rigorous analysis of multivariate calibration models was conducted based on the UV spectra in order to test its feasibility in quantifying individual protein concentrations from the mixtures. Because of the high reliability and reproducibility of UV spectra, combined with the low maintenance cost as compared to chromatography and electrophoretic techniques, this approach provided a faster, simpler, and less expensive alternative for making pooling decisions in protein purification processes.

Materials and Methods

Protein standards

Stock solutions of three different kinds of proteins, BSA, and BLG and IgG, were prepared from the purified and crystallized proteins, and used throughout this study. All proteins were purchased from Sigma-Aldrich (St. Louis, MO) and stored at a refrigerator upon arrival to the laboratory. Then, the stock solution for each protein was made by dissolving a predefined amount of the pure protein in a 0.1M phosphate

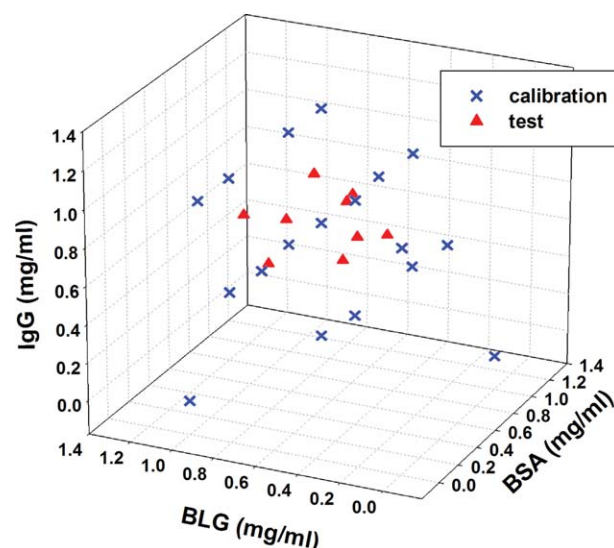


Figure 1. Concentration profiles of the ternary protein mixtures in calibration and external test set prepared by design of experiments.

buffer solution at pH 7 with a final protein concentration of 4.0 mg/ml.

Protein mixtures

Protein mixtures with known concentrations of BSA, BLG, and IgG were prepared by mixing a predefined volume of the stock protein solutions in a 0.1M phosphate buffer solution. Here, a set of sample conditions for ternary protein mixtures was generated by using central composition design with five center points and duplicate measurements²⁸ to produce a total of 37 protein samples, including three additional samples of the pure protein solution. The concentrations for the individual proteins in these samples were 0.1–1.3 mg/ml, and the range of total protein concentration was 1.2–3 mg/ml, except for the pure protein solutions, which had a concentration of 1.0 mg/ml for each protein. Then, a separate set of the protein mixture samples were additionally prepared by using a Latin Hypercube Design²⁹ to prepare a total of 10 samples, which had different protein concentrations to the previous sample set. Here, the ranges of individual protein concentrations were 0.50–0.95, 0.45–0.99, and 0.43–0.95 mg/ml for BSA, BLG, and IgG, respectively. With these two datasets, the former was used to construct the multivariate calibration model, and the latter was used as an independent test set to confirm the prediction capability of the developed calibration model, which will be further explained in the below sections. The actual concentration of the protein samples for both calibration and test sets can be seen in Figure 1.

Acquisition of UV spectra

UV absorbance spectra of the ternary protein mixtures were measured with a PerkinElmer Lambda 25 UV-VIS spectrometer (PerkinElmer, Waltham, MA). All the spectra were taken against a reference cell of 0.1M phosphate buffer solution with an optical path-length of 1 cm at room temperature. The spectral acquisition range was 250–320 nm and the spectra were recorded in an interval of 1 nm. After the spectral measurements, the data was exported in ASCII

format and imported in MATLAB (Mathworks, Natick, MA) for further analyses.

Explanatory analysis

An explanatory analysis of the UV spectra was first conducted using multivariate curve resolution alternating least squares (MCR-ALS)³⁰ and PCA.³¹ Here, MCR-ALS was utilized to resolve the spectra of the pure proteins to examine the selectivity of UV spectra in the mixture samples, and PCA was used to investigate the dominant source of variability in the given UV spectra within the principal component spaces with the reduced dimensions. In MCR-ALS, the UV spectra of the mixture samples were analyzed using the following expression of the bilinear model³⁰,

$$\mathbf{X} = \mathbf{C}\mathbf{S}^T + \mathbf{E} \quad (1)$$

where $\mathbf{X}$ is the data matrix (m -by- n) of the UV spectra, $\mathbf{C}$ is the concentration profile (m -by- k) of three individual proteins, $\mathbf{S}^T$ is the data matrix (k -by- n) of pure spectra for each component, and $\mathbf{E}$ is the residual matrix (m -by- n), respectively. In addition, m , n and k are the number of observations, wavelength variables, and pure components in the given system, respectively. In this study, the MCR-ALS model was obtained from the UV spectra of the calibration samples except for the pure protein solutions, and the model was iteratively computed using the known concentration profiles as an initial guess for $\mathbf{C}$, with non-negative constraints for both the concentration profiles and the pure spectra.³⁰ Then, pure protein spectra of BSA, BLG, and IgG were estimated by following equation,

$$\mathbf{S}^T = (\mathbf{C}^T \mathbf{C})^{-1} \mathbf{C}^T \mathbf{X} \quad (2)$$

and compared with the spectra measured experimentally using the pure protein solutions. Note that, MCR-ALS algorithm can be alternately used to calculate the concentration of the protein mixtures directly from the measured UV spectra of the mixtures and pure proteins based on follows equation.

$$\mathbf{C} = \mathbf{X}\mathbf{S}(\mathbf{S}^T \mathbf{S})^{-1} \quad (3)$$

On the other hand, in PCA model, the mean-centered UV spectra of the mixture samples including calibration and test sets were analyzed together, and the optimal number of principal components (PCs) was chosen by using the scree test based on the distribution of eigenvalues obtained from the decomposition of the spectral dataset into the lower dimensional PC space as given by,³²

$$\mathbf{X} = \mathbf{T}\mathbf{P}^T + \mathbf{E} \quad (4)$$

where $\mathbf{X}$ is the data matrix (m -by- n) of the UV spectra, $\mathbf{T}$ is the matrix (m -by- a) of scores, $\mathbf{P}$ is the matrix (n -by- a) of loads, $\mathbf{E}$ is the residual matrix (m -by- n), and a is the optimal number of PCs, respectively.

Multivariate calibration with partial least squares

PLS was employed as a multivariate calibration tool to regress the UV spectra ($\mathbf{X}$) on the protein concentrations ($\mathbf{Y}$), since it can efficiently deal with highly collinear spectral datasets based on the bilinear decomposition of $\mathbf{X}$ and $\mathbf{Y}$ into a lower dimensional latent variable space.³³ Here, PLS1 algorithm was utilized to build separate PLS models for

individual protein, and the optimal number of latent variables (LVs) in each of the PLS models was determined by leave-one-out cross-validation with the calibration set to provide highest Q^2 (≤ 1). Q^2 is a measure of predicted variance from the cross-validation, and can be used as a quantitative measure of prediction capability for the corresponding models.³¹ In addition, in this study, the different order of derivative spectra (i.e., 0th, 1st, 2nd, 3rd, and 4th order derivative) and scaling options (i.e., mean centering and auto-scaling) were employed to optimize the prediction capability of the multivariate calibration model, generating multiple PLS models for each of the three proteins. To compute the derivative spectra, Savitzky-Golay smoothing and differentiation filter was employed, and the parameters such as filter width (from 5 to 41) and polynomial order (from 1 to 5) were optimized by using the Monte-Carlo cross-validation to provide highest Q^2 . Once the optimal parameters were determined for each of the PLS models, the root mean squared error of calibration (RMSEC) and cross-validation (RMSECV) were computed from the calibration set using the following equation,

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^m (\hat{y}_i - y_i)^2}{m}} \quad (5)$$

where $\hat{y}_i$ and y_i are the estimated and measured response values for i th sample in either the model calibration or the cross-validation. Then, the root mean squared error of prediction (RMSEP) was calculated from the external test set using Eq. 5 in order to validate the prediction capability of the developed model independently. In addition, the relative error (RE) was also calculated as a complementary measure of the prediction accuracy, given by

$$\text{RE}(\%) = 100 \times \sqrt{\frac{\sum_{i=1}^m (\hat{y}_i - y_i)^2}{\sum_{i=1}^m y_i^2}} \quad (6)$$

Software

All the chemometrics analyses were done using PLS toolbox 4.0 (Eigenvector Research, Wenatchee, WA), MCR-ALS toolbox³⁰ and custom-made routines in MATLAB (Mathworks, Natick, MA).

Results and Discussion

UV spectra of pure proteins

First, the UV spectra of pure proteins, BSA, BLG, and IgG were plotted in Figure 2 in order to investigate the distinctive absorption bands for each protein. Here, the derivative spectra from zeroth (i.e., raw spectra with only smoothing) to fourth order were calculated using Savitzky-Golay filters, since they often provide better resolution for highly overlapping and broadened peaks in the UV region.³⁴ Furthermore, the absorbance signals below 260 nm in third and fourth derivative spectra were discarded throughout this study, due to the increased level of noise and artifacts created by taking the derivatives. In Figure 2a, one can see that the zeroth order spectra of three different proteins were very similar to each other, having similar absorption bands with the maximum at ~ 280 nm where the absorption coefficients followed the order of IgG > BLG > BSA. The observed peaks were originated mostly from the aromatic amino acid

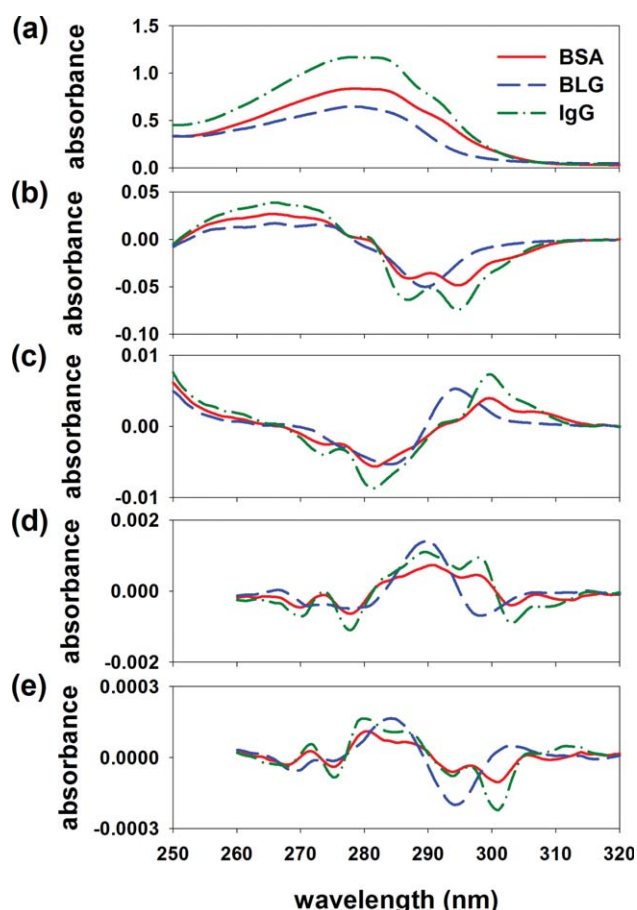


Figure 2. UV spectra of the pure protein solutions for BSA, BLG, and IgG.

(a) Zeroth order, (b) First derivative, (c) Second derivative, (d) Third derivative, and (e) Fourth derivative. All the derivative spectra including zeroth order were computed using the Savitzky–Golay smoothing and differentiation filters.

residues, such as tryptophan and tyrosine, as well as partly from the phenylalanine and disulfide bonds of cysteine.³⁵ In general, the ratio of these aromatic residues as well as the microenvironments of their chromophores for the different proteins result in slight differences in the UV absorption patterns, and these could be more clearly seen in the derivative spectra of Figures 2b–f, where three individual proteins exhibited distinctive features in the spectral region of 270–305 nm.^{36,37} Here, the UV spectrum having the most distinguished absorption patterns could be seen in BLG, exhibiting different peak wavelengths from the others. In contrast, both BSA and IgG had similar peak position at multiple wavelengths, but their molar absorptivity was significantly different from each other.

Explanatory analysis of UV spectra for protein mixtures

MCR-ALS Model of the UV Spectra. Prior to building the multivariate calibration models, the sensitivity and resolution of the UV spectra were checked quantitatively for the protein mixtures, since the UV absorption bands observed for the pure proteins of BSA, BLG, and IgG were generally overlapping to each other as illustrated in Figure 2. To this end, MCR-ALS algorithm was employed and the pure spectra of each component were mathematically resolved from

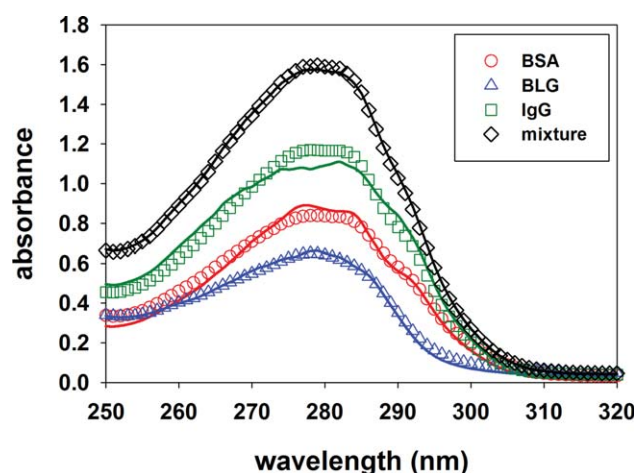


Figure 3. Pure UV spectra of the three proteins obtained from the experiments (symbols) and the MCR-ALS model (lines).

For comparison, a single spectrum of the mixture sample (0.4 mg/ml for each of the proteins) was also presented together with the mathematically resolved spectrum of the MCR-ALS model.

the UV spectral measurements and their corresponding protein concentration profiles in the calibration set. Note that, here, the three spectra of pure protein solutions were excluded during the computation of MCR-ALS model for proper comparison of them to the mathematical ones. The results obtained from this model were shown in Figure 3, where the dotted symbols and lines were used to represent the experimental and mathematically estimated spectra of the pure proteins, respectively. In this figure, one can see that MCR-ALS could correctly extract the spectral information assignable for each of the pure proteins, especially for the BLG ($r^2 = 0.995$ between the experimental and simulated spectra), which had the most distinguished UV absorption peaks as illustrated in Figure 2. Similarly, the pure experimental spectra of both the BSA and IgG were also in acceptable agreement with the spectra obtained by MCR-ALS ($r^2 = 0.978$ for BSA and $r^2 = 0.989$ for IgG) although there were slight deviations due to the highly overlapping UV absorption bands near the maximum at 280 nm, indicating that the given UV spectra can provide sufficient sensitivity for differentiating the three different proteins of BSA, BLG, and IgG in mixture states. However, it should be noted that the validation of the given MCR-ALS model based on the Eq. 3 and external test set³⁸ did not provide acceptable prediction performance except for the BLG (see Figure S2 in Supporting Information). This might be explained by the fact that the two pure spectra of BSA and IgG could not be perfectly resolved as shown in Figure 3, making the estimated concentration profiles relatively inaccurate. In this case, the use of different weighting schemes for the spectral regions, where there is a high overlap or large variance, might potentially improve the prediction accuracy of MCR-ALS model, and the variance at each data point can be used to further increase the precision by evaluating matrix **C** using variance-weighted data points.

PCA Model of the UV Spectra. As a next step, the UV spectra measured for the protein mixture samples were investigated in Figure 4a. In this figure, one can see that the spectral shape was very similar to those of pure protein

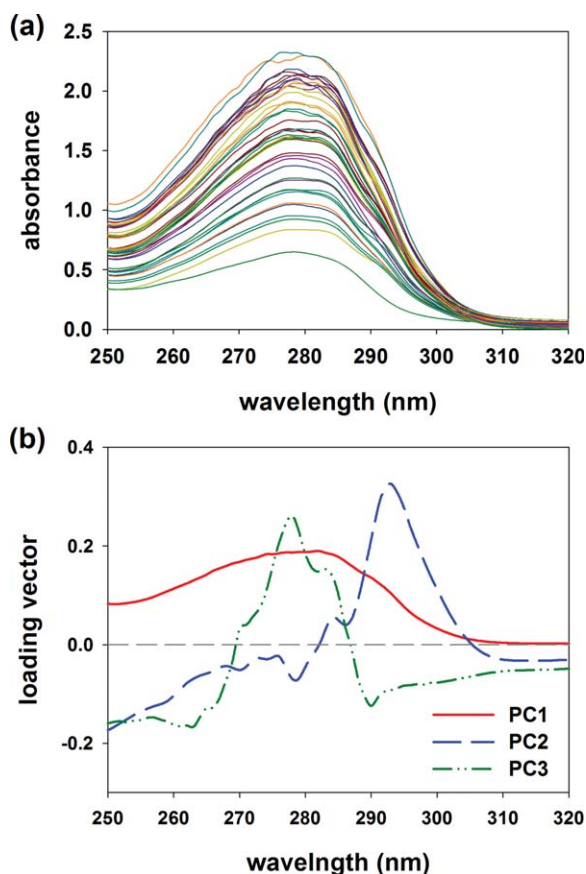


Figure 4. (a) UV spectra of the protein mixture samples. (b) The loading vectors obtained from the PCA model constructed on the given UV spectra.

samples, but they exhibited different UV absorbance at multiple wavelengths according to their protein concentration levels. At the same time, it was also observed that the spectra having the absorbance higher than 2 suffered from the increased level of noise due to the detector saturation. Therefore, after preprocessing the spectra by Savitzky-Golay smooth filter (width of 9 and polynomial order of 2), PCA model was constructed in order to analyze the important wavelength regions, which are responsible for the spectral variability seen in Figure 4a. In this PCA model, three PCs were selected as an optimum and the corresponding loading vectors were plotted in Figure 4(b), which explained 99.9 % of total variance in the data. As can be seen in this figure, first and second PCs mainly described the broad UV absorption band of tryptophan at 280 nm and the shoulder peak at around 293 nm, which might originate from tryptophan residues in proteins.³⁵ Note that the UV absorbance at 293 nm is also related with the protein conformational changes.³⁹ On the other hand, third PC generally explained the minor absorption band at around 275 nm, which might reflect the aromatic amino acid residue of tyrosine in protein.³⁵ An additional analyze revealed that the multivariate information captured by these PCs was correlated with either the concentration of three individual proteins or total protein concentration (see Figure S1 in Supporting Information), justifying the use of the multi-wavelength UV spectra employed here. However, their correlations were not strong enough to build an accurate prediction model, thus requiring the use of multivariate calibration models as will be demonstrated in the subsequent sections.

Estimation of protein concentrations using partial least squares

In order to build a reliable estimation model for the protein quantification in mixture, the multivariate calibration models based on PLS were constructed using the calibration set. During these steps, the different derivative spectra from zeroth to fourth order was employed with two different scaling options of mean-centering and auto-scaling, and their prediction performance was validated using the external test set as mentioned earlier. Note that in the external test set, one sample was found as an outlier based on the PCA model of the UV spectra (data are not shown), thus excluded throughout this study. Table 1 summarizes the results obtained from the multiple PLS models in estimating the concentration of each of the three proteins, and the models with the lowest RMSEP for each protein were represented as a bold letter. In Table 1, it is shown that the derivative spectra (i.e. from first to fourth order) gave better prediction performance, compared to the raw spectra of zeroth order, except for the IgG where the raw spectra without derivative provided equivalent prediction accuracy for the external test set. At the same time, in these models, the filter width selected by the cross-validation was relatively higher than other studies^{23,34} which typically employed 9 - 13 data points in computing the UV derivative spectra. However, in this study, the measured UV spectra had relatively low signal-to-noise ratio due to the fast scanning mode used to simulate the in-line usage, thus requiring broader filter width to compensate the enhanced level of noise through the Savitzky-Golay smoothing filter. On the contrary, regarding the scaling options, there was no clear preference to a specific one, thus their performance was dependent on several factors including the characteristics of the dataset and statistical distribution of the absorbance signals after the preprocessing.⁴⁰

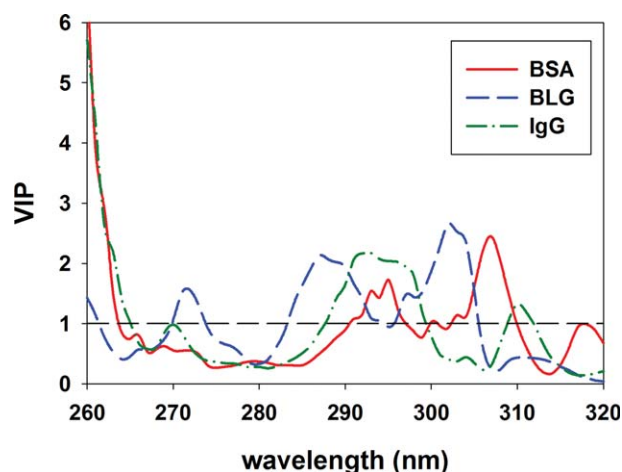
In Table 1, it could be also observed that the prediction errors of the PLS models for the BLG were relatively higher than those of BSA and IgG. This might be explained by the lower molar absorption coefficient of the BLG in the UV region,⁴¹ thus only resulting in slight improvement in its prediction compared to the MCR-ALS model (see Figure S2 in Supporting Information). However, a further analysis with variable importance in the projection (VIP) indicated that the PLS model for BLG could truly capture the important wavelength regions, differently from the other two models of BSA and IgG (see Figure S3 in Supporting Information), which was in agreement with the previous result obtained from Figure 2. Here, the VIP was used as a quantitative measure of influence for each wavelength variable in describing the covariance between **X** and **Y**, as well as in regressing **Y**, thus providing valuable information about the variable importance in the given PLS model.^{40,42} A representative VIP plot for the three different proteins with the mean-centered third derivative spectra can be seen in Figure 5, where the three different PLS models for BSA, BLG and IgG emphasized the different UV regions distinctly in their own models, demonstrating the sensitivity and specificity of the UV spectra in estimating the proteins in mixtures.

Finally, when it comes to the best prediction performance among the various PLS models, the mean-centered third derivative, auto-scaled fourth derivative, and mean-centered third derivative spectra exhibited the most accurate prediction power for the BSA, BLG and IgG, respectively. In these models, the prediction accuracy of RMSEP was 0.036, 0.088 and 0.049 for the BSA, BLG and IgG with their relative error (RE) of 4.8, 12.0 and 6.8 %, providing the satisfactory

Table 1. Summary of Performance for the Multivariate Calibration Models Developed from the Different Derivative Spectra and the Scaling Options

			Parameter*	# of LVs	RMSEC†	RMSECV	RMSEP†
BSA	0th	Mean-centering	17 (3)	7	0.052 (7.0)	0.069	0.076 (10.2)
		Auto-scaling	25 (5)	7	0.038 (5.2)	0.056	0.081 (10.9)
	1st	Mean-centering	19 (3)	6	0.055 (7.4)	0.065	0.075 (10.1)
		Auto-scaling	37 (5)	4	0.053 (7.2)	0.068	0.057 (7.7)
	2nd	Mean-centering	17 (2)	8	0.049 (6.6)	0.080	0.058 (7.8)
		Auto-scaling	19 (2)	5	0.054 (7.3)	0.071	0.057 (7.7)
	3rd	Mean-centering	33 (3)	7	0.055 (7.4)	0.077	0.036 (4.8)
		Auto-scaling	33 (5)	8	0.050 (6.7)	0.076	0.061 (8.3)
	4th	Mean-centering	39 (4)	7	0.056 (7.5)	0.103	0.049 (6.7)
		Auto-scaling	37 (4)	8	0.047 (6.3)	0.079	0.037 (5.0)
BLG	0th	Mean-centering	27 (5)	7	0.038 (5.1)	0.057	0.115 (15.6)
		Auto-scaling	21 (1)	4	0.047 (6.3)	0.054	0.119 (16.1)
	1st	Mean-centering	29 (5)	8	0.038 (5.1)	0.065	0.126 (17.1)
		Auto-scaling	21 (2)	5	0.053 (7.1)	0.070	0.091 (12.3)
	2nd	Mean-centering	29 (5)	7	0.046 (6.2)	0.083	0.094 (12.8)
		Auto-scaling	21 (2)	7	0.038 (5.1)	0.071	0.105 (14.3)
	3rd	Mean-centering	35 (3)	8	0.036 (4.8)	0.077	0.101 (13.7)
		Auto-scaling	27 (3)	8	0.039 (5.2)	0.073	0.101 (13.7)
	4th	Mean-centering	37 (4)	7	0.043 (5.8)	0.093	0.124 (16.8)
		Auto-scaling	33 (5)	8	0.038 (5.1)	0.079	0.088 (12.0)
IgG	0th	Mean-centering	25 (4)	6	0.055 (7.4)	0.067	0.063 (8.8)
		Auto-scaling	5 (3)	4	0.052 (7.0)	0.061	0.049 (6.9)
	1st	Mean-centering	11 (1)	5	0.061 (8.1)	0.069	0.076 (10.7)
		Auto-scaling	19 (2)	5	0.066 (8.8)	0.079	0.072 (10.2)
	2nd	Mean-centering	17 (2)	5	0.059 (7.9)	0.090	0.067 (9.5)
		Auto-scaling	23 (2)	6	0.053 (7.1)	0.070	0.088 (12.3)
	3rd	Mean-centering	39 (3)	6	0.056 (7.5)	0.083	0.049 (6.8)
		Auto-scaling	35 (4)	4	0.061 (8.1)	0.071	0.073 (10.4)
	4th	Mean-centering	41 (4)	6	0.048 (6.5)	0.069	0.090 (12.6)
		Auto-scaling	27 (5)	4	0.056 (7.5)	0.082	0.088 (12.4)

*Values indicate the filter width and polynomial order (in parenthesis) used in Savitzky-Golay filter. †Values in parenthesis represent the relative error (RE) in %.

**Figure 5. VIP scores calculated from the PLS models estimating the concentration of BSA, BLG and IgG.**

In these PLS models, the third derivative spectra were used with the mean-centering.

estimation capability for the three proteins investigated here. This could be again confirmed in Figure 6 where the actual vs. predicted protein concentrations were plotted. In Figure 6, the estimated protein concentrations from the UV spectra were generally in good agreement with the actual ones. In order to further validate the estimated prediction power of the corresponding PLS models, a rigorous analysis with permutation test was also conducted for these models^{19,31} (see Fig. S4 in Supporting Information), and all the three model showed no symptoms of over-fitting or abnormality, thus demonstrating the validity of the multivariate calibration model in quantifying the protein mixtures from the UV spectra.

In summary, the prediction accuracy of the developed PLS model for quantifying the protein mixtures was acceptable owing to the use of multi-wavelength UV spectra, and their validity was rigorously analyzed using the various kinds of chemometrics tools. In addition, it is worth to note that the approach employed here was different from the previous studies,^{24,26} where a single type of mixture design has been employed without considering the use of fully independent test set. Table 2 summarized the advantages of the proposed approach in terms of chromatographic operation for protein purification, compared to other methodologies reported in the literature.^{14–16,35} Its fast, cheap and simple nature poses a great potential to implement it in the protein purification processes as a real-time quantitative monitoring tool to improve the process efficiency and consistency. However, in order to take the real benefits of this approach in the protein purification, there still exist many practical issues and challenges to be solved; (i) interference of the UV absorbance from the high impurities in the harvested samples, (ii) possible changes in the protein conformation within the chromatography columns, (iii) the diverse range of concentrations for the different proteins and under the varying operational conditions, and (iv) integration of the real-time UV spectroscopy in the chromatographic system. Therefore, our next logical work will be the validation of this approach in the real-time operation of the chromatography system, which is currently on-going.

Conclusion

In this study, we tested the feasibility of using the multi-wavelength UV spectra in estimating the protein

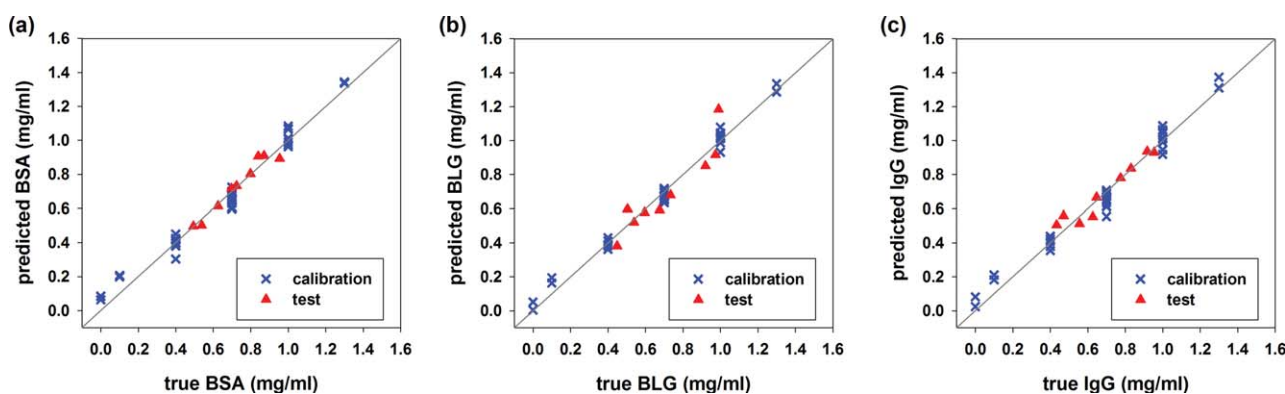


Figure 6. Actual versus predicted values of the protein concentrations obtained from the best PLS models for the calibration and external test sets.

(a) BSA, (b) BLG, and (c) IgG.

Table 2. Comparison of Different Approaches in Real-Time Quantitative Monitoring of the Proteins in the Chromatographic Protein Purification

Approach	Summary	Pros	Cons	Analysis Time
Offline HPLC analysis of column fractions	Used when high resolution separation is necessary to estimate CQA	Base of CQA. Reliable and robust	Time consuming and expensive. No real time data	30 min
Pooling by monitoring absorbance at 280 nm	Very common in biotech processes and current chromatographic instruments	Simple to implement	Incapable of estimating the composition of multicomponent mixtures	Real time
Online HPLC for real time pooling	Samples are withdrawn from the eluent and analyzed in real time	Offers real time product quality attributes information	Requires slowing down flow rate to allow for HPLC analysis time	13.5 min
Use of UPLC for real time pooling	High pressure faster liquid chromatography method	Faster analysis than traditional HPLC	Expensive due to high pressure operations	~8 min
At-line Fluorescence analysis	Use of tryptophan fluorescence for estimating CQA of target protein	Faster than traditional HPLC and simpler to operate	Offline analysis increases complexity. Impractical when species differ in tryptophan accessibility	~20 min
Online full range UV spectroscopy	Full range UV spectra method for estimating composition of protein mixtures	Provides real time data. UV probe can be easily introduced in effluent stream. Non-destructive	New and unexploited. Requires engineering different probe sizes for different columns.	30 s

concentrations in mixture as a real-time monitoring tool for the protein purification processes. Three different proteins of BSA, BLG, and IgG having different molecular weights and chemical structures were used as a model system, and various kinds of chemometrics tools were employed to verify the prediction capability of the protein quantification models from the UV spectra of the protein mixtures. Both the explanatory and predictive analysis revealed that the UV spectra of the given ternary protein mixture had sufficient resolution and sensitivity to estimate the concentrations of the different proteins, and the optimal quantification models could be successfully identified from the PLS regression models with various preprocessing techniques. These results suggest that this approach can be used as a viable tool in making the real-time pooling decisions in the chromatographic operations of protein purification, at least for the given mixture of proteins investigated here. To further generalize these observations, more studies will be needed under the conditions of different protein mixtures and concentration ranges, as well as with the real chromatographic samples, which will be our next work. Note that, however, this approach is analogous to the existing hyphenated HPLC with multiwavelength diode-array detector (HPLC-DAD), which

has been developed to improve the resolution and fingerprinting capability of the resultant chromatogram.⁴³ Therefore, we expect that this approach can be readily applicable to the various kinds of protein purification processes in order to ensure the stability and consistency of the protein production in biopharmaceutical manufacturing or other related field.

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